CS304P - Quiz No.04

Date: 08 July 2025

Q.1 Which of the following is true about friend functions in template classes?

A. They can only access public members of the class.
B. They must be defined inside the class.
C. They are inherited by derived classes.
D. They can access private and protected members of the class.

Q.2 What keyword is used to define a generic function template?

A. typename
B. generic
C. template
D. class

Q.3 If a function template has parameters `T` and `U`, calling `func(1, 2.5)` results in:

A. Both deduced as double
B. Compilation error
C. T=int, U=double
D. Both deduced as int

Q.4 When is a class template instantiated?

A. When linked
B. When declared
C. At runtime
D. At compile time

Q.5 Which of the following is the correct syntax for creating a template class X friend of a template class Y?

A. X<T> friend;
B. X<T>;
C. friend X;
D. friend X<T>;

Q.6 What feature enables same-named template functions to exist?

A. Specialization
B. Function overloading
C. Inheritance
D. Polymorphism

Q.7 Which keyword is used to define a template in C++?

A. class
B. function
C. typename
D. template

Q.8 To accept two type parameters, _____ is written.

A. template<class T, U>
B. template<typename T U>
C. template<class T U>
D. template<typename T, typename U>

Q.9 Which of the following is the correct syntax for creating a function Test(T) friend of a template class?

A. void Test(T);
B. void friend Test(T);
C. void Test(T) friend;
D. friend void Test(T);

Q.10 Which of the following is an example of partial specialization?

A. template<class T, float>
B. template<int>
C. template<int , float , char>
D. template<class T, class U>
